



AI Solution Intern - Case Evaluation Submission



Instructions

- Submit a **PDF** (max 20 MB) via the form at the bottom of this page.
- **12–15 pages** (excluding appendix).
- **Deadline:** 1 week from the date you receive this case.

The AI Solution Intern Case — AI-Powered Career Companion

Background

CareerUp is an HR Tech startup that aims to help **soon-to-graduate students** find jobs, apply effectively, and prepare themselves to land the right position.

Currently, fresh graduates face several challenges:

- **Not knowing what jobs suit them** — lacking personalized self-assessment and career guidance
- **Not knowing which companies are a good fit** — lacking visibility into what companies are hiring, what their culture and values are like, and whether they

align with the student's goals and interests

- **Writing unattractive resumes/CVs** — not knowing how to highlight their strengths in a way that matches the position they are applying for
- **Preparing for interviews ineffectively** — having no mock interviews tailored to specific positions and companies
- **Scattered information** — having to search across multiple sources (job boards, LinkedIn, company reviews, salary data)

The company wants to build an **AI Career Companion** that helps **personalize the entire job search process** for fresh graduates — from self-discovery to landing a job.

Your challenge: Analyze the problems fresh graduates face at each stage of the job search process, then design a solution that leverages **AI, automation tools, SaaS platforms, APIs, or any combination of available technologies** to solve those problems in a personalized way. You decide what the system should do and which tools to use — but you must be specific and justify your choices.

Your Task

Design an **AI Career Companion** that personalizes the entire job search process for fresh graduates, divided into **2 parts**:

Part 1: Problem Analysis & Solution Design

1. **Pain Point Analysis** — Analyze the problems fresh graduates face during the job search process. What obstacles exist at each stage? And how can technology help solve each of these problems?
2. **Proposed Features** — Based on your analysis, propose the features the system should have. Explain what problem each feature solves and why you chose it.
3. **Personalization Strategy** — Explain how the system will use the chosen technologies to personalize recommendations for each individual user.
4. **System Architecture & Tech Stack** — Present a high-level system diagram and choose the technologies you would use, along with your reasoning.

5. **AI & Tool Selection** — Identify the **specific AI models, APIs, frameworks, and tools** you would use (e.g., GPT-4o, Claude, LangChain, Pinecone, n8n, Notion, etc.).

Part 2: Detailed Solution Design

From the features you proposed in Part 1, select **1–2 of the most important features** and provide a **detailed solution design**:

- A **high-level system architecture diagram** showing how the components connect (user-facing layer, backend, AI services, data stores, external APIs, etc.)
- A **detailed walkthrough** of how the selected features work end-to-end — describe the data flow, how the chosen tools and services are integrated, and how the system personalizes the output for each user
- Clearly show how the **specific tools and technologies** you selected in Part 1 fit into the architecture
- What **limitations or risks** exist in your design and how would you address them?

▼ *Example — What an answer in Part 2 looks like (different use case: Trip Planning)*

Below is a **simplified example** using a **different domain** (trip planning) to illustrate the expected scope and depth. Your submission should follow a similar structure but applied to the career companion case.

Selected Feature: Personalized Itinerary Generator

Pain point it solves: Travelers spend hours researching across Google Maps, blogs, TripAdvisor, and airline sites to piece together an itinerary that fits their budget, interests, and schedule.

Architecture Diagram (high-level):

Personalized Itinerary Generator

|
└─ Airtable Interface – user-facing layer where user input preferences,

```

|                                     browse itineraries, and edit
their trip plans
|
|— n8n (automation hub)
|   |— Claude API — generates personalized day-by
-day itinerary from user preferences
|   |— Google Places API — fetches real-time venu
e data, ratings, hours, photos
|   |— Skyscanner API — pulls flight & hotel pric
ing, generates booking deep links
|   |— OpenWeatherMap API — fetches weather forec
ast for pre-trip email
|   |— Scheduled triggers — sends pre-trip checkl
ist emails 3 days before departure
|
|— Airtable (data layer) — stores trip data, user pre
ferences, itinerary history

```

End-to-end walkthrough:

1. User fills in a trip request via **Airtable Interface** — destination, travel dates, budget, and interests (e.g., "Chiang Mai, 4 days, \$15k budget, food & temples"). This creates a new record in Airtable.
2. New record triggers an **n8n workflow** → sends a structured prompt to **Claude API** including user preferences and constraints → receives a draft itinerary.
3. n8n enriches each suggested place via **Google Places API** — pulling ratings, opening hours, and photos to validate the suggestion is real and currently open.
4. n8n calls **Skyscanner API** to fetch flight and hotel options within budget → selects best-fit options and generates **deep links** (affiliate URLs) to book directly on the provider's site (e.g., Skyscanner, [Booking.com](#), Agoda).
5. The completed itinerary — including booking links — is written back to **Airtable** → user reviews, adjusts, and clicks through to book directly on provider sites via the **Airtable Interface**.

6. 3 days before the trip, a scheduled **n8n workflow** triggers an email with a packing checklist, weather forecast (via OpenWeatherMap API), and a PDF itinerary.

Why these tools:

- *Airtable (Interface + data layer)* — serves as both the user-facing frontend and the database. No custom frontend or separate DB needed — Airtable Interfaces let users input, browse, and edit data directly, while the underlying tables store all trip and preference data.
- *n8n* — orchestrates the entire workflow (API calls, data enrichment, scheduled emails) without writing backend code. Connects all services in one place.
- *Claude API* — strong at generating structured, context-aware text; handles multi-constraint planning well.
- *Google Places API* — ensures suggestions are grounded in real, up-to-date venue data (not hallucinated).
- *Skyscanner API* — aggregates pricing across providers and generates deep links for direct booking, saving users from manual comparison. No need to build payment processing — users book on the provider's site.

Tool & API references:

Tool	Documentation / Reference
Airtable (Interface + API)	airtable.com/developers/web/api/introduction
n8n (automation)	docs.n8n.io
Claude API (Anthropic)	docs.anthropic.com/en/api
Google Places API	developers.google.com/maps/documentation/places/web-service
Skyscanner API (via RapidAPI)	developers.skyscanner.net/docs/intro
OpenWeatherMap API	openweathermap.org/api

Limitations & risks:

- *AI hallucination* — Claude may suggest places that don't exist → mitigated by validating every suggestion against Google Places API.
 - *API rate limits & cost* — Google Places and Skyscanner have usage caps → implement caching and batch requests.
 - *Stale data* — venue hours or prices may change → show "last updated" timestamps and allow user refresh.
 - *No in-app booking* — the system redirects users to provider sites for booking rather than handling payments directly → avoids payment gateway complexity and PCI compliance, but means the user leaves the platform briefly to complete the transaction.
-

Deliverables Checklist

☐ **PDF Document** containing:

- ☐ Pain Point Analysis
 - ☐ Proposed Features with rationale
 - ☐ Personalization Strategy
 - ☐ System Architecture & Tech Stack
 - ☐ AI & Tool Selection with justification
 - ☐ High-level architecture diagram & detailed feature walkthrough for 1–2 key features
 - ☐ Limitations & risks with mitigation strategies
-

Submit your work

<https://www.notion.so/edf76722484b4590966d61f39033e3f6?pvs=112>